

Issues with the healthcare supply chain

Supply chain management (SCM) can be defined as the integration of business processes from end users through manufacturers. It provides products, services and information, apart from adding value for customers.

In healthcare, supply chain management is a very long, complex and fragmented process. Companies are expanding their product portfolios to meet rapidly changing markets and shorter product life cycles. Emerging economies want more affordable products. Quality and compliance issues are rising because products are more complex and regulatory scrutiny is strict. Moreover, increasing drug recalls is also a huge issue for companies as it is hard to track the distribution. The many challenges that this sector faces can be read at <http://thefintechtimes.com/future-healthcare-blockchain>.

The whole supply process is very fragmented in a large country like India. Procuring raw materials, managing supplies and delivering goods to providers and patients is a complex process. Also, the information regarding medical products and services usually passes through many hands including manufacturers, agencies, insurance companies, hospitals and providers, giving rise to privacy concerns.

Hence, there's an increased chance of information getting falsified somewhere in between. Any forged documents or tampered supplies can cost companies a large amount of money and pose a danger to patients' health. Therefore, increasing the efficiency and transparency of the healthcare supply chain becomes critical for manufacturers.

Existing mechanisms in the healthcare supply chain

In a developing country like India, each clinic manages its own inventory and sales data. Here, the problem is that the store does not know what its future requirements might be. Nobody knows what problem the next patient will bring, or which epidemic will break out. Currently, everything operates based on

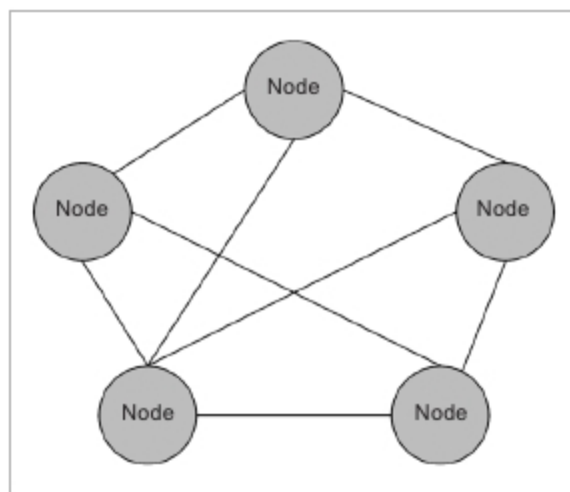


Figure 5: Decentralised database — nodes contact the nearest node that has the required data instead of any central access point

the experience of the nurses and other staff. After this, an online purchase order (PO) is placed, which is an ID given to the order to be used for future references. This PO is included as an identity mark for any item that is part of this purchase order, bringing down processing time. For every further step, this PO is passed as a reference, thus maintaining a global reference number for that consignment.

First, after receiving the PO, the wholesaler checks the stock and if available, the status is set to be delivered within five days. If not, the wholesaler contacts the supplier and then the clinic has to wait longer for the supplies to arrive. This situation arises due to the delay in conveying information about the requirements, as the supply chain is not managed efficiently.

Second, the packaging and delivery of products is also done based on the order received through the PO. A delivery form has the contents of the order. This is paper-intensive work and any tampering can lead to a number of unwanted events; the obvious being money loss, spurious drugs and illegal trading.

Third, during the transportation of medicines, there is no check except for the driver of the truck. Such lack of oversight can be exploited to deliver counterfeit medicines. This can be prevented if medicines are tagged using bar codes, which can be stored on the blockchain for later inspection. The healthcare supply chain is so complex that no one can guarantee whether or not the medicine comes from the right manufacturer.

Also, when patients have unique requirements, calling for special products to be designed and delivered on an urgent basis, the logistical complexities ensure that it might take days for the product to even go into the design process.

Moreover, as reports, prescriptions and other patient related materials are issued by the staff members of the hospital, the documents can be easily lost or damaged. In other cases, the same patient can go to a number of doctors getting the same medicine prescribed repeatedly for the same health condition. Also, getting prescribed medicines from different chemists is fairly easy. This drug abuse takes many lives every year.

Most of the documents are always filled by the staff and not the doctor. Thus, these documents can be easily forged and tampered with, and are not reliable at all. This lack of trust is a big bottleneck when patients need to get reimbursements from insurance companies, quickly.

Advantages of integrating the blockchain in healthcare

Every time a product changes hands, the transaction could be documented electronically, creating a permanent history of it, from manufacturing to point of sale. Smart contracts at every stage can carry out transactions that needed a third party earlier, after the terms of the contracts are met. This could dramatically reduce time delays, added costs, and human error that plague the system today.

Some other advantages of using the blockchain in supply chain

Linking: Physical goods can be linked to bar codes, digital tags and serial numbers for better tracking, or can be linked to each other and used as needed.

Assigning: Verifying and assigning if a product needs some special treatment in transit—for example, if it needs refrigeration.

Tracking: Purchase orders, change orders, receipts, prescriptions and everything involved can be saved digitally and tracked on the blockchain. Also, the blockchain ensures immutability and better interpolation of data.